

SAVANNAH HILTON HEAD INTL, GA, USA

WMO: 722070

Lat: 32.131N

Lon: 81.202W

Elev: 14

StdP: 101.16

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03822

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.5	(c) -0.7	(d) -11.7	(e) 1.4	(f) 3.3	(g) -9.1	(h) 1.7	(i) 4.2	(j) 10.6	(k) 14.0	(l) 9.2	(m) 11.6	(n) 2.0	(o) 310	(p) 0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.9	(c) 35.3	(d) 25.1	(e) 34.1	(f) 24.9	(g) 33.0	(h) 24.7	(i) 26.8	(j) 31.9	(k) 26.3	(l) 31.2	(m) 25.8	(n) 30.5	(o) 3.5	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 25.6	(b) 20.8	(c) 28.6	(d) 25.1	(e) 20.2	(f) 28.1	(g) 24.6	(h) 19.6	(i) 27.8	(j) 83.6	(k) 31.8	(l) 81.4	(m) 31.3	(n) 79.4	(o) 30.6	(p) 29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.5	(b) 7.6	(c) 6.8	(e) -5.6	(f) 37.3	(g) 1.9	(h) 1.3	(i) -7.0	(j) 38.3	(k) -8.1	(l) 39.0	(m) -9.2	(n) 39.7	(o) -10.6	(p) 40.7		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	19.6	10.2	12.2	15.4	19.2	23.3	26.6	28.0	27.6	25.2	20.3	14.9	11.8	(6)
(7)		DBStd	7.33	5.30	4.90	4.56	3.81	3.05	2.13	1.55	1.63	2.38	3.90	4.45	5.12	(7)
(8)		HDD10.0	152	64	29	10	1	0	0	0	0	0	0	9	39	(8)
(9)		HDD18.3	938	255	178	112	35	3	0	0	0	1	26	120	208	(9)
(10)		CDD10.0	3660	71	91	178	277	411	498	558	545	456	321	157	97	(10)
(11)		CDD18.3	1404	3	7	21	62	156	248	300	287	207	88	18	7	(11)
(12)		CDH23.3	12684	15	47	164	513	1359	2341	3068	2799	1653	593	101	31	(12)
(13)		CDH26.7	4989	1	4	26	129	493	1016	1398	1189	587	136	8	1	(13)
(14)	Wind	WSAvg	2.9	3.1	3.2	3.5	3.3	3.1	2.8	2.6	2.5	2.8	2.7	2.7	2.8	(14)
(15)	Precipitation	PrecAvg	1253	92	77	98	80	95	153	166	173	117	74	54	69	(15)
(16)		PrecMax	1859	228	201	229	268	222	366	510	433	310	504	134	147	(16)
(17)		PrecMin	804	11	7	2	10	9	25	34	26	9	0	4	3	(17)
(18)		PrecStd	240	56	43	60	53	54	83	95	94	73	90	38	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	27.3	29.2	31.4	34.9	36.5	36.7	36.3	34.2	31.4	27.8	25.9	(19)
MCWB			19.0	19.2	18.8	20.6	22.7	24.8	25.3	26.2	23.9	23.9	21.3	20.0	(20)	
2%		DB	22.9	24.8	27.0	29.8	32.7	34.7	35.2	34.6	32.5	29.5	26.0	24.1	(21)	
		MCWB	17.4	17.7	18.3	20.0	22.4	24.6	25.2	25.7	24.0	22.4	20.0	19.0	(22)	
5%		DB	20.7	22.6	25.2	28.2	31.1	33.3	34.0	33.3	31.3	28.2	24.3	22.0	(23)	
		MCWB	16.6	16.5	17.4	19.6	22.2	24.4	25.3	25.3	23.8	21.6	18.7	18.4	(24)	
10%		DB	18.7	20.3	23.2	26.5	29.5	31.8	32.7	32.1	30.1	26.8	22.6	20.1	(25)	
		MCWB	15.4	15.3	16.6	18.8	21.6	24.0	25.2	25.1	23.5	20.9	18.5	17.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.5	20.6	21.2	23.0	25.3	26.9	27.5	27.7	26.5	25.7	23.3	21.5	(27)
MCDWB			23.0	24.8	25.4	27.5	30.4	32.9	32.4	33.0	29.8	28.5	25.2	24.0	(28)	
2%		WB	19.0	19.3	20.1	21.9	24.4	26.0	26.8	26.8	25.7	24.5	21.8	20.3	(29)	
		MCDWB	21.4	23.1	24.1	26.8	29.1	31.5	32.0	31.7	29.1	27.3	23.8	22.4	(30)	
5%		WB	17.7	18.0	19.1	21.1	23.6	25.5	26.2	26.3	25.2	23.6	20.6	19.1	(31)	
		MCDWB	20.0	21.1	22.8	25.6	28.0	30.7	31.6	31.0	28.7	26.2	23.1	21.5	(32)	
10%		WB	16.0	16.6	18.0	20.2	22.8	24.9	25.7	25.8	24.7	22.7	19.3	17.7	(33)	
		MCDWB	18.3	19.4	21.5	24.4	27.1	29.8	30.7	30.2	28.0	25.2	21.8	20.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.6	12.0	12.2	12.3	11.4	10.1	9.9	9.3	9.2	10.9	12.0	11.3	(35)
MCDBR			12.9	13.6	13.7	13.2	12.9	11.9	11.2	10.8	10.5	11.2	12.7	12.0	(36)	
MCWBR			8.2	7.7	6.5	5.5	4.8	4.1	3.8	3.7	3.7	5.2	7.2	7.6	(37)	
5% WB		MCDWB	11.2	11.8	11.0	11.2	10.8	10.3	10.1	9.7	8.5	8.7	9.9	10.7	(38)	
		MCWBR	8.3	7.7	6.2	5.6	4.6	4.1	3.9	3.7	3.5	4.6	6.7	7.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.329	0.333	0.367	0.403	0.452	0.496	0.528	0.512	0.429	0.396	0.356	0.345	(40)	
taud		2.525	2.515	2.440	2.331	2.227	2.156	2.080	2.126	2.356	2.415	2.479	2.504	(41)		
Ebn at Noon		894	926	913	888	843	802	775	782	840	844	853	853	(42)		
Edh at Noon		85	95	110	127	142	152	164	154	118	104	89	84	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2.96	3.59	4.71	5.93	6.38	6.33	6.22	5.63	4.78	4.16	3.27	2.61	(44)	
(45)		RadStd	0.16	0.36	0.42	0.39	0.39	0.48	0.39	0.35	0.40	0.44	0.22	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.59	+0.79	N/A	N/A	N/A	N/A	N/A	-83	+216	+151
(47) Regional (35 neighbors)		+0.57	+0.79	N/A	N/A	N/A	N/A	N/A	-91	+193	+131

Nomenclature: See separate page